Ordinary Differential Equation

- **Ordinary Differential Equations (ODE)** relate a function $y(t)$ to its derivatives.
- An **Autonomous** ODE has no explicit dependence on t.
- The order of the highest derivative in an ODE is the **Order**.
- A first order **Initial Value Problem (IVP)** is an ODE with an **Initial Condition (IC)** $y(t_0) = y_0$.
- A function satisfying the IC and the ODE for $a < t_0 < b$ solves the IVP on (a, b).

Example: Solution of a First-Order IVP

Solve the IVP $y' = 2y$, $y(0) = 3$.

The solution of the IVP is $\boxed{y(t) = 3e^{2t}}$

Check: $y'(t) = 6e^{2t} = 2y(t)$ and $y(0) = 3$.

Second Order to First Order

1. Start with a second-order IVP

$$y'' = g(t, y, y'), \qquad y(t_0) = a_1, \qquad y'(t_0) = a_2$$

2. Define $y_1 = y$ and $y_2 = y'$. Then

$$\begin{aligned} y_1' &= y_2 & y_1(t_0) &= a_1 \\ y_2' &= g(t, y_1, y_2) & y_2(t_0) &= a_2 \end{aligned}$$

3. With $\boldsymbol{y} = (y_1, y_2)$ and $\boldsymbol{y}_0 = (a_1, a_2)$, write

$$\boldsymbol{y}' = \boldsymbol{f}(t, \boldsymbol{y}), \qquad \boldsymbol{y}(t_0) = \boldsymbol{y}_0, \qquad \boldsymbol{f}(t, \boldsymbol{y}) = \begin{pmatrix} y_2 \\ g(t, y_1, y_2) \end{pmatrix}$$

Mass and Spring

Example: Second Order to First Order

- A mass $m > 0$, damping coefficient $c \geq 0$, spring stiffness $k > 0$, and applied force $g(t)$ give

$$my'' + cy' + ky = g(t)$$

- Displacement $y_1 = y$ and velocity $y_2 = y'$ give

$$\underbrace{\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}'}_{\boldsymbol{y}'} = \underbrace{\begin{pmatrix} y_2 \\ \big(g(t) - cy_2 - ky_1\big)/m \end{pmatrix}}_{\boldsymbol{f}(t,\boldsymbol{y})} \iff \boxed{\boldsymbol{y}' = \boldsymbol{f}(t, \boldsymbol{y})}$$

- Initial displacement and velocity supply the two initial values.

Existence and Uniqueness

First-order IVP

If $\boldsymbol{f}$ and its Jacobian $D_{\boldsymbol{y}}\boldsymbol{f}$ (with entries $\partial f_i/\partial y_j$) are continuous near $(t_0, \boldsymbol{y}_0)$, then

$$\boldsymbol{y}' = \boldsymbol{f}(t, \boldsymbol{y}), \qquad \boldsymbol{y}(t_0) = \boldsymbol{y}_0$$

has a unique local solution.

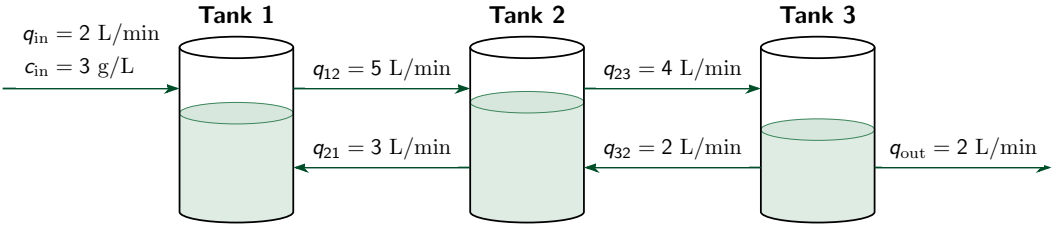
nth-order IVP

If g and its partials wrt $y, y', \ldots, y^{(n-1)}$ are continuous near the IC , then the IVP

$$y^{(n)} = g(t, y, y', \ldots, y^{(n-1)}), \qquad y^{(j)}(t_0) = a_j \quad (j = 0, \ldots, n-1)$$

has a unique local solution.

Well Mixed Tanks



Tank Volumes

1. Volume change in L/min is inflows minus outflows.

$$v_1' = 2 + 3 - 5 = 0$$
$$v_2' = 5 + 2 - 3 - 4 = 0$$
$$v_3' = 4 - 2 - 2 = 0$$

2. Volumes remain constant. $v_1 = 60\ \mathrm{L}, \quad v_2 = 80\ \mathrm{L}, \quad v_3 = 50\ \mathrm{L}$
3. Well mixed means concentration in g/L is

$$c_i(t) = y_i(t)/v_i$$

where y_i is the amount of salt in grams and v_i is the volume in liters.

Salt Flow Rate

1. A volume flow (q L/min) at concentration (c g/L) carries salt at qc g/min.

$$q\,c: \qquad \frac{\cancel{\text{L}}}{\text{min}}\frac{\text{g}}{\cancel{\text{L}}} = \frac{\text{g}}{\text{min}}$$

2. For example, the inflow brings salt into tank 1 at

$$\left(2\,\frac{\cancel{\text{L}}}{\text{min}}\right)\left(3\,\frac{\text{g}}{\cancel{\text{L}}}\right) = 6\,\frac{\text{g}}{\text{min}}$$

3. Flow from tank 1 to tank 2 carries salt at the rate

$$q_{12}c_1(t) = 5c_1(t) = 5y_1(t)/60 = y_1(t)/12 \text{ g/min}$$

Salt Balances

1. Salt mass change in $\mathrm{g/min}$ is inflows minus outflows.

$$y_1' = 2(3) + 3c_2 - 5c_1 = 6 + 3y_2/80 - 5y_1/60$$
$$y_2' = 5c_1 + 2c_3 - (3+4)c_2 = 5y_1/60 + 2y_3/50 - 7y_2/80$$
$$y_3' = 4c_2 - (2+2)c_3 = 4y_2/80 - 4y_3/50$$

2. All concentrations are in $\mathrm{g/L}$. All volume flow rates are in $\mathrm{L/min}$.

Initial Salt Masses

1. Salt mass in g is volume (v L) times concentration (c $\mathrm{g/L}$)

$$y_i(0) = v_i(0)c_i(0), \qquad \cancel{\mathrm{L}}\,\frac{\mathrm{g}}{\cancel{\mathrm{L}}} = \mathrm{g}$$

2. The initial salt masses are

$$y_1(0) = (60\,\cancel{\mathrm{L}})(2\,\mathrm{g}/\cancel{\mathrm{L}}) = 120\,\mathrm{g}$$
$$y_2(0) = (80\,\cancel{\mathrm{L}})(1\,\mathrm{g}/\cancel{\mathrm{L}}) = 80\,\mathrm{g}$$
$$y_3(0) = (50\,\cancel{\mathrm{L}})(0\,\mathrm{g}/\cancel{\mathrm{L}}) = 0\,\mathrm{g}$$

Tank System as a Vector IVP

- The salt mass vector $\boldsymbol{y} = (y_1, y_2, y_3)$ has units g.
- Collect terms

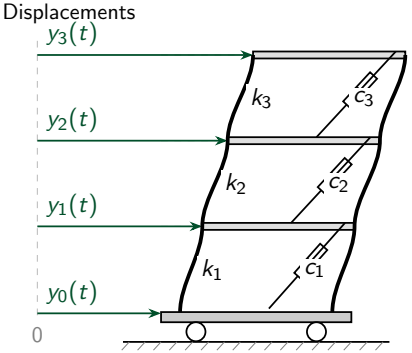
$$\boxed{\boldsymbol{y}' = \boldsymbol{A}\boldsymbol{y} + \boldsymbol{b}}, \qquad \boldsymbol{y}(0) = \begin{pmatrix} 120 \\ 80 \\ 0 \end{pmatrix}$$

$$\boldsymbol{A} = \begin{pmatrix} -1/12 & 3/80 & 0 \\ 1/12 & -7/80 & 1/25 \\ 0 & 1/20 & -2/25 \end{pmatrix}, \qquad \boldsymbol{b} = \begin{pmatrix} 6 \\ 0 \\ 0 \end{pmatrix}$$

- Entries of $\boldsymbol{A}$ have units $1/\mathrm{min}$. Entries of $\boldsymbol{b}$ have units g/min.
- Internal transfers cancel in the total salt balance

$$(y_1 + y_2 + y_3)' = 6 - 2c_3 = 6 - y_3/25$$

Three-Story Building



$m_3 = 2\,\mathrm{kg}$
$k_3 = 2\,\mathrm{N/m}$
$c_3 = 1\,\mathrm{N\,s/m}$

$m_2 = 3\,\mathrm{kg}$
$k_2 = 3\,\mathrm{N/m}$
$c_2 = 1\,\mathrm{N\,s/m}$

$m_1 = 4\,\mathrm{kg}$
$k_1 = 4\,\mathrm{N/m}$
$c_1 = 2\,\mathrm{N\,s/m}$

Prescribed base motion $y_0(t)$

Story Forces

1. Rightward displacement is positive. Floor displacement $y_i(t)$ and prescribed base motion $y_0(t)$ are in m.
2. Relative displacement and velocity in story i are

$$y_i - y_{i-1} \quad [\mathrm{m}], \qquad y_i' - y_{i-1}' \quad [\mathrm{m/s}]$$

3. Transmitted force (s_i N) is spring force plus damping force

$$s_i = k_i(y_i - y_{i-1}) + c_i(y_i' - y_{i-1}')$$

4. Story i exerts $-s_i$ on its upper floor and $+s_i$ on the level below.

Force Units

1. Spring force in N is stiffness (k_i $\mathrm{N/m}$) times relative displacement ($y_i - y_{i-1}$ m)

$$k_i(y_i - y_{i-1}): \qquad \frac{\mathrm{N}}{\cancel{\mathrm{m}}}\cancel{\mathrm{m}} = \mathrm{N}$$

2. Damping force in N is damping (c_i $\mathrm{N\,s/m}$) times relative velocity ($y_i' - y_{i-1}'$ $\mathrm{m/s}$)

$$c_i(y_i' - y_{i-1}'): \qquad \frac{\mathrm{N}\,\cancel{\mathrm{s}}}{\cancel{\mathrm{m}}}\frac{\cancel{\mathrm{m}}}{\cancel{\mathrm{s}}} = \mathrm{N}$$

3. Force in N is mass (m_i kg) times acceleration (y_i'' $\mathrm{m/s^2}$)

$$m_i y_i'': \qquad \mathrm{kg}\,\frac{\mathrm{m}}{\mathrm{s}^2} = \mathrm{N}$$

Floor Balances

1. Net force in N is mass times acceleration

$$4y_1'' = -s_1 + s_2, \qquad 3y_2'' = -s_2 + s_3, \qquad 2y_3'' = -s_3$$

2. Use the displayed stiffness and damping values

$$\begin{aligned} s_1 &= 4(y_1 - y_0) + 2(y_1' - y_0') \\ s_2 &= 3(y_2 - y_1) + (y_2' - y_1') \\ s_3 &= 2(y_3 - y_2) + (y_3' - y_2') \end{aligned}$$

3. Collect floor terms on the left and base terms on the right

$$\begin{aligned} 4y_1'' + 3y_1' - y_2' + 7y_1 - 3y_2 &= 2y_0' + 4y_0 \\ 3y_2'' - y_1' + 2y_2' - y_3' - 3y_1 + 5y_2 - 2y_3 &= 0 \\ 2y_3'' - y_2' + y_3' - 2y_2 + 2y_3 &= 0 \end{aligned}$$

Second-Order Matrix System

- With $\boldsymbol{y} = (y_1, y_2, y_3)$, the floor balances are

$$\boxed{\boldsymbol{M}\boldsymbol{y}'' + \boldsymbol{C}\boldsymbol{y}' + \boldsymbol{K}\boldsymbol{y} = \boldsymbol{r}(t)}$$

$$\boldsymbol{M} = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{pmatrix}, \quad \boldsymbol{C} = \begin{pmatrix} 3 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix}, \quad \boldsymbol{K} = \begin{pmatrix} 7 & -3 & 0 \\ -3 & 5 & -2 \\ 0 & -2 & 2 \end{pmatrix}$$

$$\boldsymbol{r}(t) = \begin{pmatrix} 2y_0'(t) + 4y_0(t) \\ 0 \\ 0 \end{pmatrix}$$

- Entries of $\boldsymbol{M}$, $\boldsymbol{C}$, and $\boldsymbol{K}$ have units kg, $\mathrm{N\,s/m}$, and $\mathrm{N/m}$. The force vector $\boldsymbol{r}$ has units N.

Six First-Order Equations

1. Velocity $\boldsymbol{v} = \boldsymbol{y}'$ has units $\mathrm{m/s}$.
2. Solve the second-order system for acceleration

$$\boldsymbol{y}' = \boldsymbol{v}, \qquad \boldsymbol{v}' = -\boldsymbol{M}^{-1}\boldsymbol{K}\boldsymbol{y} - \boldsymbol{M}^{-1}\boldsymbol{C}\boldsymbol{v} + \boldsymbol{M}^{-1}\boldsymbol{r}(t)$$

3. Stack position and velocity into a six-component state

$$\boldsymbol{z} = \begin{pmatrix} \boldsymbol{y} \\ \boldsymbol{v} \end{pmatrix}, \qquad \boxed{\boldsymbol{z}' = \begin{pmatrix} \boldsymbol{0} & \boldsymbol{I} \\ -\boldsymbol{M}^{-1}\boldsymbol{K} & -\boldsymbol{M}^{-1}\boldsymbol{C} \end{pmatrix} \boldsymbol{z} + \begin{pmatrix} \boldsymbol{0} \\ \boldsymbol{M}^{-1}\boldsymbol{r}(t) \end{pmatrix}}$$

Here $\boldsymbol{0}$ and $\boldsymbol{I}$ are 3×3 matrices.

Acceleration Units

- Acceleration in $\mathrm{m/s^2}$ is force (N) divided by mass (kg)

$$\frac{\mathrm{N}}{\mathrm{kg}} = \frac{\cancel{\mathrm{kg}}\,\mathrm{m}}{\mathrm{s}^2\,\cancel{\mathrm{kg}}} = \frac{\mathrm{m}}{\mathrm{s}^2}$$

- Every term in $\boldsymbol{v}' = -\boldsymbol{M}^{-1}\boldsymbol{K}\boldsymbol{y} - \boldsymbol{M}^{-1}\boldsymbol{C}\boldsymbol{v} + \boldsymbol{M}^{-1}\boldsymbol{r}$ has these units

$$\underbrace{\frac{1}{\cancel{\mathrm{kg}}}\frac{\cancel{\mathrm{kg}}}{\mathrm{s}^2}\,\mathrm{m}}_{\boldsymbol{M}^{-1}\boldsymbol{K}\boldsymbol{y}} = \frac{\mathrm{m}}{\mathrm{s}^2}, \qquad \underbrace{\frac{1}{\cancel{\mathrm{kg}}}\frac{\cancel{\mathrm{kg}}}{\mathrm{s}}\frac{\mathrm{m}}{\mathrm{s}}}_{\boldsymbol{M}^{-1}\boldsymbol{C}\boldsymbol{v}} = \frac{\mathrm{m}}{\mathrm{s}^2}$$

First-Order Building IVP

$$\boldsymbol{z}' = \begin{pmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ -7/4 & 3/4 & 0 & -3/4 & 1/4 & 0 \\ 1 & -5/3 & 2/3 & 1/3 & -2/3 & 1/3 \\ 0 & 1 & -1 & 0 & 1/2 & -1/2 \end{pmatrix} \boldsymbol{z} + \begin{pmatrix} 0 \\ 0 \\ 0 \\ y_0 + y_0'/2 \\ 0 \\ 0 \end{pmatrix}$$

- Initial positions ($\boldsymbol{a}$ m) and velocities ($\boldsymbol{b}$ $\mathrm{m/s}$) give

$$\boldsymbol{z}(0) = \begin{pmatrix} \boldsymbol{a} \\ \boldsymbol{b} \end{pmatrix}$$

- Prescribe $y_0(t)$ and these six initial values to complete the IVP.